Math 10C — Unit 6: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

---

1. Determine the length of each line segment.

(a) $A(3, 5)$ to $B(9, 5)$

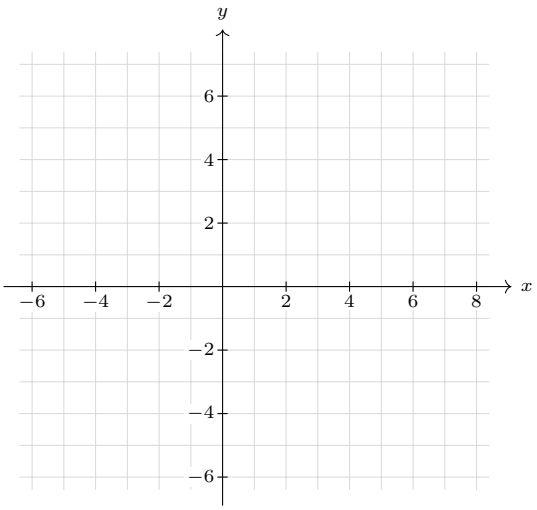
(b) $C(-2, 4)$ to $D(-2, 15)$

(c) $K(9, -14)$ to $L(-33, -14)$

(d) $P(6187, -1054)$ to $Q(6187, 975)$

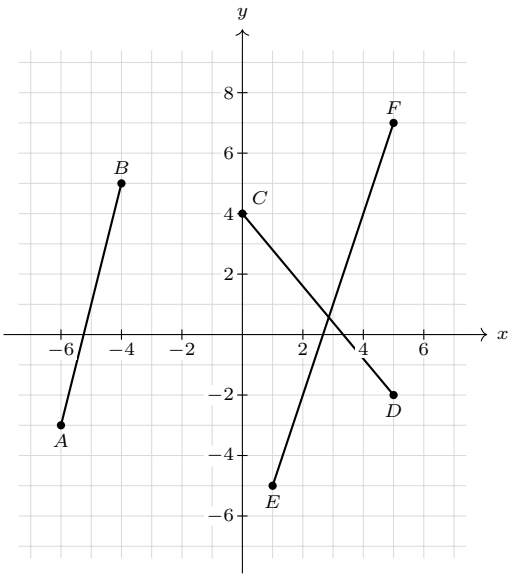
2. A triangle has vertices $P(-5, -4)$, $Q(7, -4)$, and $R(2, 6)$.

- Sketch the triangle on the grid.
- Calculate the area of the triangle.



3. Use the Pythagorean theorem to determine the length of each segment shown on the grid. Give each answer as **(i)** an exact value in simplest radical form, and **(ii)** a decimal to the nearest hundredth.

(a) Segment AB



(b) Segment CD

(c) Segment EF

4. Determine the distance between each pair of points.

(a) $A(1, -3)$ and $B(9, 3)$

(b) $C(-1, 4)$ and $D(11, 9)$

(c) $E(1, -1)$ and $F(7, 1)$, as an exact value in simplest radical form

(d) $P(5, -1)$ and $Q(-3, 6)$, to the nearest hundredth

5. After school, Maria walks 5 blocks east and 8 blocks north to reach her house from the school. Nathan walks 7 blocks west and 4 blocks south to reach his house from the same school.

(a) Taking the school as the origin, list the coordinates of each house.

(b) If a block represents 110 metres, determine the direct distance, to the nearest metre, between the two houses.

6. Determine the coordinates of the midpoint of the line segment with the given pair of endpoints.

(a) $A(3, 5),\ C(9, 13)$

(b) $X(-4, -9),\ Y(-12, 3)$

(c) $K(11, -15),\ L(-9, 7)$

(d) $P(-3.5, 4.2),\ Q(2.5, -7.8)$

7. In each case, M is the midpoint of $\overline{AB}$. Determine the value of x.

(a) $A(3, 7),\ B(9, x),\ M(6, 2)$

(b) $A(5, 8),\ B(x, 0),\ M(0, 4)$

8. A cell tower is to be built halfway between the rural towns of Ashford and Birchwood, with a service road connecting both towns to the tower. On a Cartesian plane, Ashford is located at $(2148, 1876)$ and Birchwood is located at $(3562, 2394)$.

(a) What are the coordinates of the tower?

(b) Determine the length of the service road, to the nearest kilometre, if the grid scale is 1 unit represents 50 metres.

9. *(Multiple Choice)* Horizontal line segment CD, with endpoints $C(4, -2)$ and $D(x, y)$, has a length of 6 units. Which of the following must be true?

A. $x = 10$ B. $y = 4$ or -8 C. $x + y = 6$ D. $x = y$ or $x - y = 12$

10. *(Multiple Choice)* The distance between the points $(3, -2)$ and $(7, 1)$ is

A. 5 B. $\sqrt{12}$ C. 7 D. 25

11. *(Numerical Response)* To the nearest tenth, the perimeter of $\triangle PQR$ with vertices $P(5, 9)$, $Q(5, 1)$, and $R(-2, 9)$ is _____.

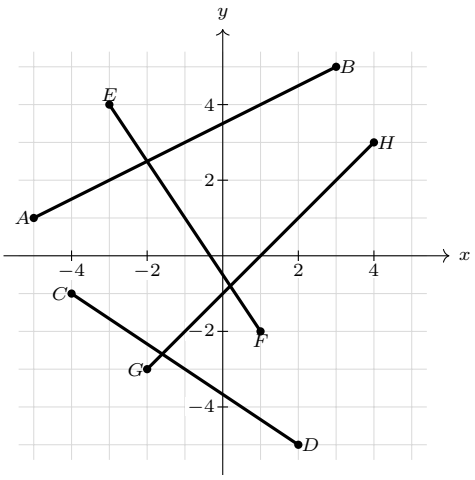
Record your answer left to right. | | | | |

12. *(Numerical Response)* The midpoint of line segment ST is $M(2, -4)$. If the coordinates of T are $(-5, 3)$, and the coordinates of S are (x, y), the value of x is _____.

Record your answer left to right. | | | | |

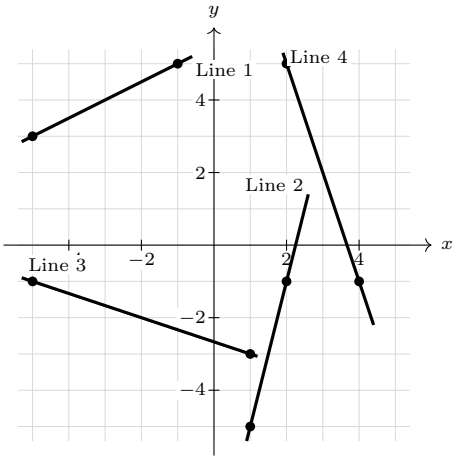
Lesson 2 Assignment *Show your work.*

1. Each line segment on the grid has endpoints with integer coordinates. Complete the table.



Segment	Rise	Run	Slope
AB			
CD			
EF			
GH			

2. Each of the lines below passes through the two marked points, which have integer coordinates. Calculate the slope of each line.

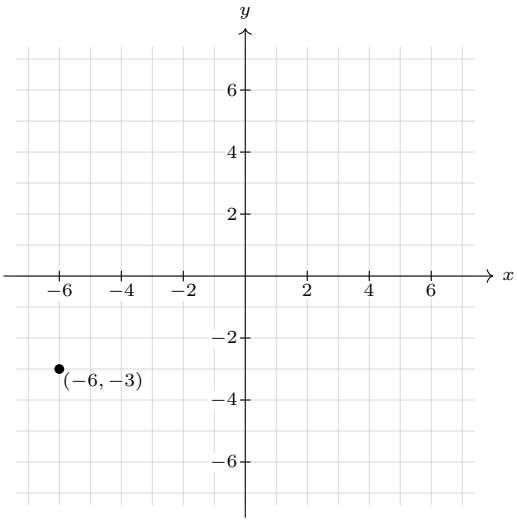


slope of Line 1 = slope of Line 3 =

slope of Line 2 = slope of Line 4 =

3. Draw a line segment on the grid which passes through the point $(-6, -3)$ and has a slope of $\frac{2}{3}$. The segment must be long enough to cross both the x-axis and the y-axis. Write the coordinates of three other points on the segment which have integer coordinates.

Three other points with integer coordinates:



4. For each slope and point below, plot the line through the given point and list three other points on it that have integer coordinates. (The line does not have to cross both axes.) Draw both lines of a pair on the same grid.

(a) slope $= \frac{3}{5}$, $(3, -1)$

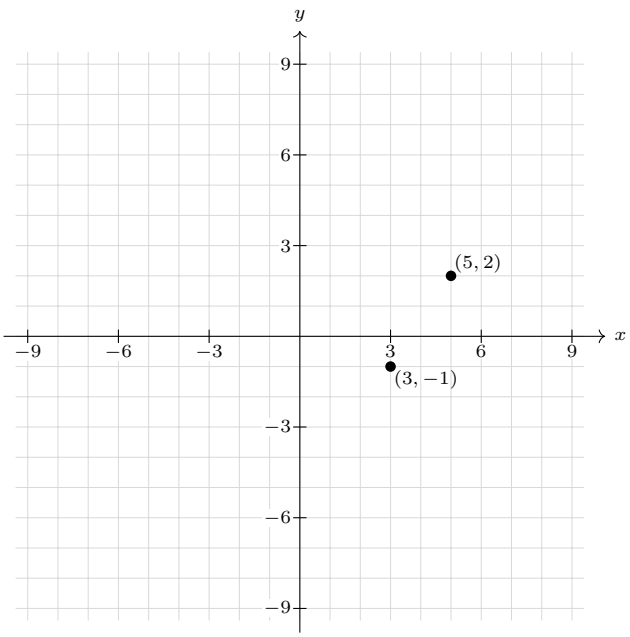
(b) slope $= -\frac{1}{4}$, $(5, 2)$

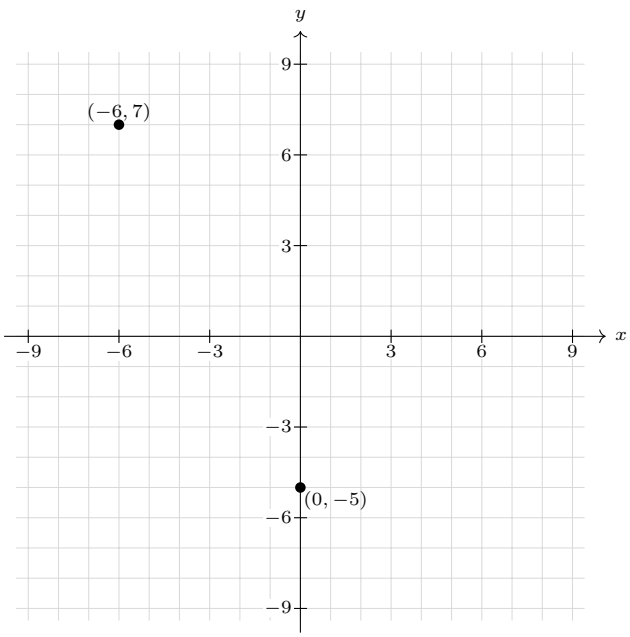
(c) slope $= -\frac{5}{2}$, $(-6, 7)$

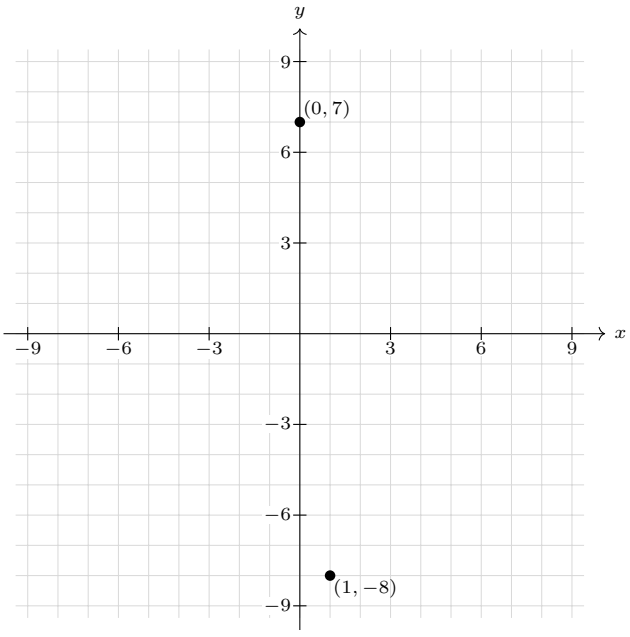
(d) slope $= 3$, $(0, -5)$

(e) slope $= -3$, $(1, -8)$

(f) slope $= 0$, $(0, 7)$







Three other points with integer coordinates:

(a) _____

(b) _____

(c) _____

(d) _____

(e) _____

(f) _____

5. P has coordinates $(2, -3)$. Find two positions for point Q so that the slope of PQ is

(a) 3 (b) -2 (c) $\frac{2}{5}$

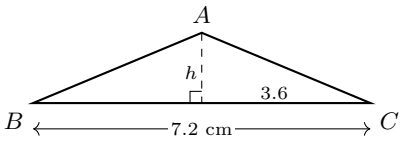
(d) $-\frac{3}{4}$ (e) 0 (f) undefined

6. Two of the three measures rise, run, and slope are given below. Calculate the value of the third measure in each case.

(a) slope $= \frac{3}{8}$, run $= 40$ (b) slope $= -\frac{2}{7}$, rise $= -8$ (c) slope $= -\frac{5}{12}$, run $= 36$ (d) slope $= \frac{7}{10}$, rise $= 21$

7. A skateboard ramp has a slope of $\frac{3}{5}$. Calculate the height of the ramp if its base length is 2.5 metres.

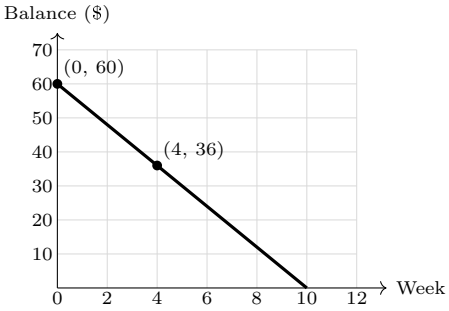
8. Triangle ABC is isosceles with $AB = AC$ and $BC = 7.2$ cm. Calculate the area of the triangle if the slope of $AC = -\frac{5}{12}$. (The height h meets BC at its midpoint.)



9. A cyclist rides at a steady pace. Her distance–time graph passes through $(0, 0)$ and (2.5, 45), where time is in hours and distance is in kilometres.

(a) Determine the slope of the graph, including its units. (b) Explain what the slope tells you about the cyclist.

10. The graph shows the balance on Jaden's bus pass during the first weeks of school.



(a) Determine the slope of the graph, including its units.

(b) What does the slope mean in this situation?

(c) After how many weeks will the pass be empty?

11. A highway sign warns of an 8% downhill grade.

(a) Write 8% as a fraction in lowest terms.

(b) How many metres does the road drop over a horizontal distance of 1.5 km?

12. A building code requires a wheelchair ramp to have a slope no greater than $\frac{1}{12}$. A ramp must rise 45 cm to reach a doorway.

(a) A contractor plans a ramp with a run of 4.8 m. Does it meet the code? Show how you know.

(b) What is the shortest run, in metres, that meets the code?

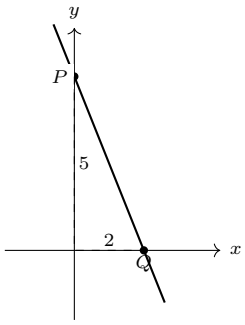
13. *(Multiple Choice)* The slope of $\overline{PQ}$ is

A. $\frac{2}{5}$

B. $-\frac{2}{5}$

C. $\frac{5}{2}$

D. $-\frac{5}{2}$



14. *(Multiple Choice)* The point $(-3, 0)$ is on a line which has a slope of $-\frac{3}{4}$. The next point with integer coordinates on the line, to the right of $(-3, 0)$, is

A. $(1, -3)$

B. $(1, 3)$

C. $(-7, 3)$

D. $(-7, -3)$

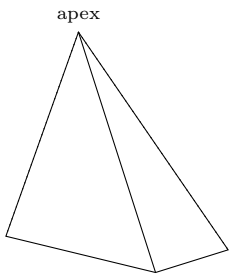
15. *(Multiple Choice)* W is a point in quadrant I, X is a point in quadrant II, Y is a point in quadrant III, and Z is a point in quadrant IV. Which one of the following statements must be true?

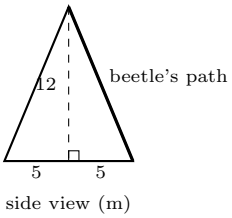
A. Line segment WX has a positive slope.
B. Line segment XY has a positive slope.
C. Line segment WY has a positive slope.
D. Line segment XZ has a positive slope.

16. *(Multiple Choice)* Four lines have the slopes listed below. Which line is the steepest?

A. $m = 3$ B. $m = -4$ C. $m = \frac{1}{5}$ D. $m = -\frac{3}{2}$

Use the following information to answer questions 17 and 18. A pyramid has a square base of side length 10 m and a vertical height of 12 m.





side view (m)

17. *(Numerical Response)* A beetle starts at the midpoint of one edge of the base and climbs straight up the face to the apex, as in the side view. To the nearest tenth, the slope of the beetle's climb is _____.

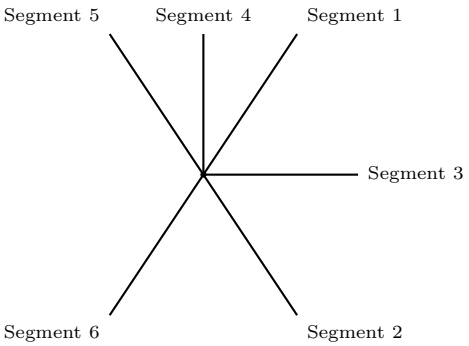
Record your answer left to right. ☐☐☐☐

18. *(Numerical Response)* A fly starts at a corner of the base and climbs the pyramid along the slanted edge to the apex. To the nearest tenth, the slope of the fly's climb is _____.

Record your answer left to right. ☐☐☐☐

Lesson 3 Assignment *Show your work.*

1. Each labelled segment below starts at the centre dot. State whether the slope of each segment is positive, negative, zero, or undefined.



Segment 1: _____
Segment 2: _____
Segment 3: _____
Segment 4: _____
Segment 5: _____
Segment 6: _____

2. Use the slope formula to calculate the slope of the line segment with the given endpoints.

(a) $A(10, -3)$ and $B(-2, 5)$

(b) $C(-5, 2)$ and $D(3, -6)$

(c) $P(-15, -4)$ and $Q(-3, -13)$

(d) $S(14, -33)$ and $T(-8, -15)$

(e) $U(-50, -20)$ and $V(-50, 40)$

(f) $K(9, -37)$ and $L(384, -37)$

3. A coordinate grid is superimposed on the cross-section of a ski hill. The coordinates of the bottom and the top of a straight path up the hill are, respectively, $(2, 5)$ and $(20, 68)$, where the units are metres.

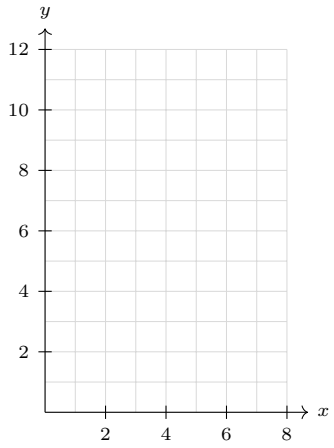
(a) Calculate the slope of the path.

(b) Calculate the coordinates of the midpoint of the path.

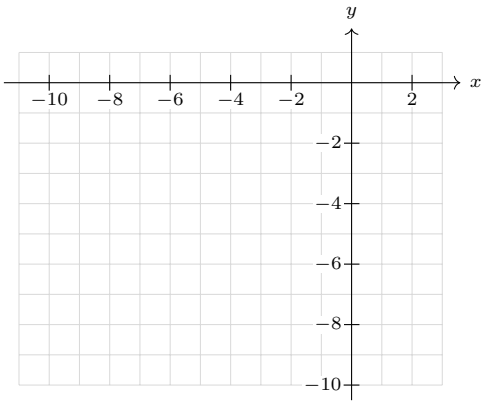
(c) Calculate the length of the path, to the nearest tenth of a metre.

4. The line segment joining each pair of points below has the given slope. Determine the value of k in each case, and draw the line segment on the accompanying grid.

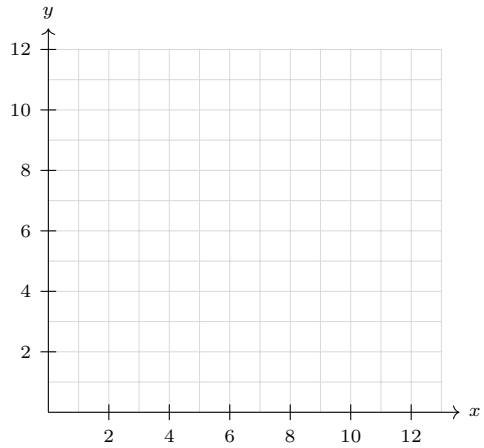
(a) $S(3,5)$, $T(6,k)$; $m=2$



(b) $L(k,-5)$, $M(2,-9)$; $m=-\frac{1}{3}$



(c) $U(4,7)$, $V(k,10)$; $m=\frac{3}{8}$



5. Consider points $X(-10,15)$, $Y(-3,12)$ and $Z(11,6)$.

(a) Use the slope formula to prove that X, Y and Z are collinear.

(b) Use the distance formula to prove that X, Y and Z are collinear.

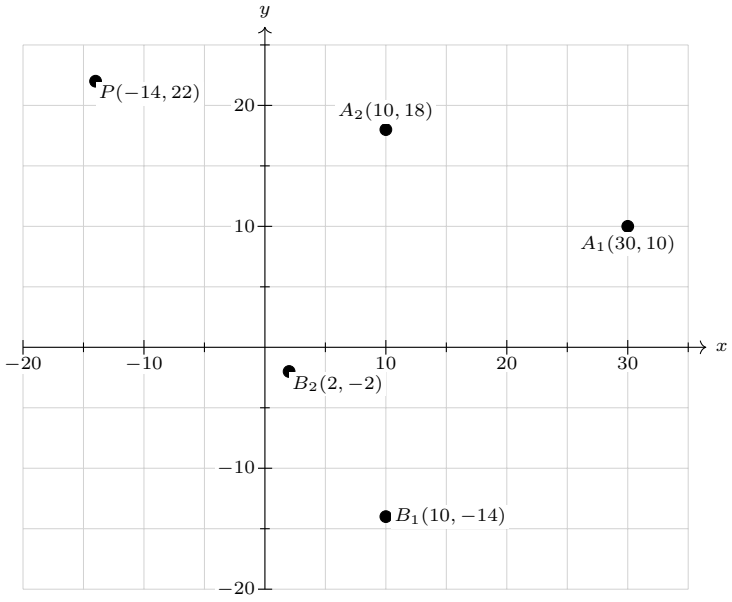
6. Consider points $A(-10,-1)$, $B(2,5)$ and $C(18,13)$.

(a) Prove that A, B and C are collinear.

(b) Does the point $D(6,9)$ lie on line segment AC? Explain.

(c) Find the value of k if the point $E(k,k)$ lies on line segment AC.

7. A hiking party's base camp is lost in the wilderness at the point $P(-14, 22)$. A rescue crew sets out by truck from $A_1(30, 10)$, driving in a straight line through $A_2(10, 18)$. A second crew sets out by boat from $B_1(10, -14)$, travelling in a straight line through $B_2(2, -2)$. If both crews continue travelling in these directions, will either of them reach the base camp? Explain using slopes.



8. *(Multiple Choice)* The slope of the line segment joining $E(6, -2)$ and $F(2, 10)$ is

A. -4 B. $\frac{1}{3}$ C. -3 D. $-\frac{1}{3}$

9. *(Multiple Choice)* If the line segment joining $(5, 4)$ and $(11, k)$ has slope $-\frac{5}{6}$, then $k =$

A. -9 B. -1 C. 9 D. 13

10. *(Multiple Choice)* One endpoint of a line segment is $(2, 9)$. The other endpoint lies on the x-axis. If the slope of the segment is $-\frac{3}{2}$, then the midpoint of the segment is

A. $(5, 4.5)$ B. $(10, 4.5)$ C. $(5, 9)$ D. $\left(\frac{1}{2}, 5\right)$

11. *(Numerical Response)* $D(3, -7)$, $E(11, 15)$ and $F(19, 4)$ are the vertices of a triangle. The slope of the median from D to the midpoint of EF, to the nearest tenth, is _____.

Record your answer left to right. | | | | |

12. The slopes of two line segments, AB and PQ, are given. Determine whether the segments are parallel, perpendicular, or neither.

(a) $m_{AB} = \frac{6}{16},\ m_{PQ} = \frac{3}{8}$ (b) $m_{AB} = \frac{5}{4},\ m_{PQ} = -\frac{4}{5}$ (c) $m_{AB} = \frac{2}{7},\ m_{PQ} = \frac{4}{14}$

(d) $m_{AB} = \frac{9}{10},\ m_{PQ} = \frac{10}{9}$ (e) $m_{AB} = -6,\ m_{PQ} = \frac{1}{6}$ (f) $m_{AB} = \frac{3}{5},\ m_{PQ} = \frac{5}{3}$

13. The slopes of eight line segments are given. Which pairs of segments are parallel, and which pairs are perpendicular?

$m_{AB} = 8$ $m_{CD} = -\frac{1}{8}$ $m_{EF} = -3$ $m_{GH} = 8$ $m_{IJ} = \frac{1}{3}$ $m_{KL} = -\frac{1}{2}$ $m_{MN} = 2$ $m_{PQ} = \frac{1}{3}$

14. Is the segment from $(2, 1)$ to $(2, 7)$ perpendicular to the segment from $(-3, 4)$ to $(5, 4)$? Explain.

15. A, B and C are the points $(1, 6)$, $(-5, -2)$ and $(7, 2)$ respectively. Determine the slope of a line

(a) parallel to AB (b) perpendicular to AB (c) parallel to BC (d) perpendicular to AC

16. The vertices of two triangles are given. Determine if either triangle is right-angled.

(a) $\triangle PQR \rightarrow P(-2, 4),\ Q(0, 0),\ R(-6, -2)$ (b) $\triangle ABC \rightarrow A(-5, 7),\ B(1, 9),\ C(3, 3)$

17. In each case, the slopes of two parallel lines are given. Determine the value of the variable.

(a) $5,\ \frac{k}{4}$ (b) $-3,\ \frac{6}{n}$ (c) $\frac{7}{9},\ 4m$ (d) $\frac{5}{6},\ -\frac{w}{12}$

18. In each case, the slopes of two perpendicular lines are given. Determine the value of the variable.

(a) $\frac{1}{4},\ 4h$ (b) $6,\ \frac{12}{p}$ (c) $-4,\ \frac{s}{3}$ (d) $-\frac{2}{5},\ -\frac{q}{10}$

19. $P(-6,-2)$ and $R(3,4)$ are opposite vertices of a rhombus $PQRS$. Find the slope of the other diagonal, QS. (Recall: the diagonals of a rhombus are perpendicular.)

20. A, B and C are the points $(-4,2)$, $(0,7)$ and $(5,3)$ respectively.

(a) Use the slope formula to show that $\triangle ABC$ is right-angled at B.

(b) Use the distance formula to confirm your result in part (a) with the Pythagorean theorem.

21. *(Multiple Choice)* A and B are the points $(3,5)$ and $(-1,7)$ respectively. A line perpendicular to AB will have slope

A. -2 B. $-\frac{1}{2}$ C. $\frac{1}{2}$ D. 2

22. The line segment joining $U(-5,p)$ and $V(-2,3)$ is perpendicular to the line segment joining $X(1,6)$ and $Y(7,2)$. Find the value of p, to the nearest tenth.

23. Dev calculated the slope of the segment from $M(-2,7)$ to $N(4,-5)$ like this:

$$m_{MN} = \frac{-5-7}{-2-4} = \frac{-12}{-6} = 2.$$

He then said that a line perpendicular to MN has slope $-\frac{1}{2}$.

(a) Identify and explain Dev's error.

(b) Determine the correct slope of MN and of a line perpendicular to MN.

24. *(Numerical Response)* A taxi's fare increases at a constant rate with the distance driven. A 4 km trip costs \$12.10 and a 10 km trip costs \$26.50. The slope of the graph of fare (in dollars) against distance (in kilometres), in dollars per kilometre, is _____.

Record your answer left to right. | | | | |

25. Consider the points $A(-3, k)$, $B(1, 2)$ and $C(9, 8)$.

(a) Find the value of k so that A, B and C are collinear.

(b) The segment from $D(2, -1)$ to $E(5, t)$ is perpendicular to line AC. Find the value of t.

Lesson 4 Assignment *Show your work.*

1. Decide whether each table represents a linear relation. If it does, state the slope.

(a)

x	1	2	3	4	5
y	4	7	10	13	16

(b)

x	0	2	4	6	8
y	10	9	7	4	0

(c)

x	-3	-1	1	3	5
y	2	2	2	2	2

(d)

x	0	1	3	6	10
y	5	3	-1	-7	-15

2. Complete each table so that it represents a linear relation.

(a)

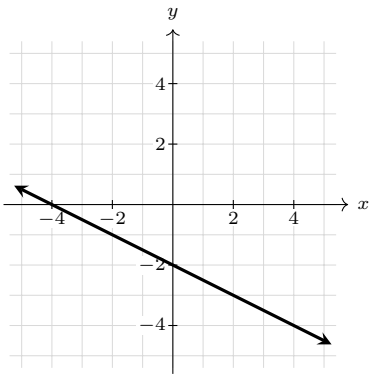
x	0	1	2	3	4	5
y	7	4			-5	

(b)

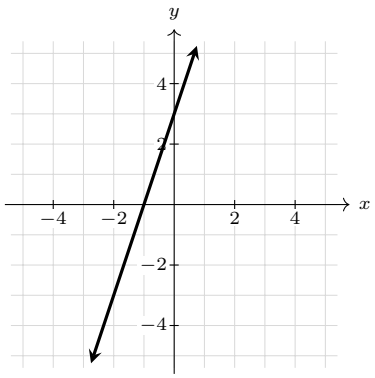
x	-4	-2	0	2	4	6
y		0	1		3	

3. For each line, state the x-intercept, the y-intercept and the slope.

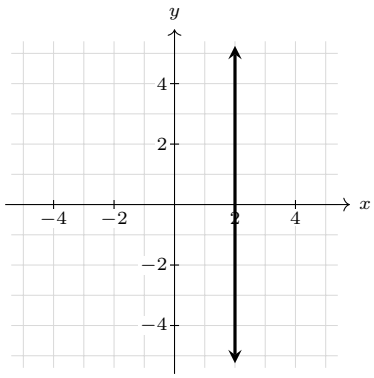
(a)



(b)



(c)



4. Each table represents a linear relation. State the x-intercept and the y-intercept.

(a)

x	-6	-3	0	3	6
y	-4	-2	0	2	4

(b)

x	-2	0	2	4	6
y	10	6	2	-2	-6

5. State the domain and range of each linear relation.

(a) The line through $(1, 2)$ and $(3, 6)$, extending forever in both directions.

(b) The horizontal line through $(-3, 5)$.

(c) The vertical line through $(4, -1)$.

(d) The line segment from $(-2, 7)$ to $(3, -3)$.

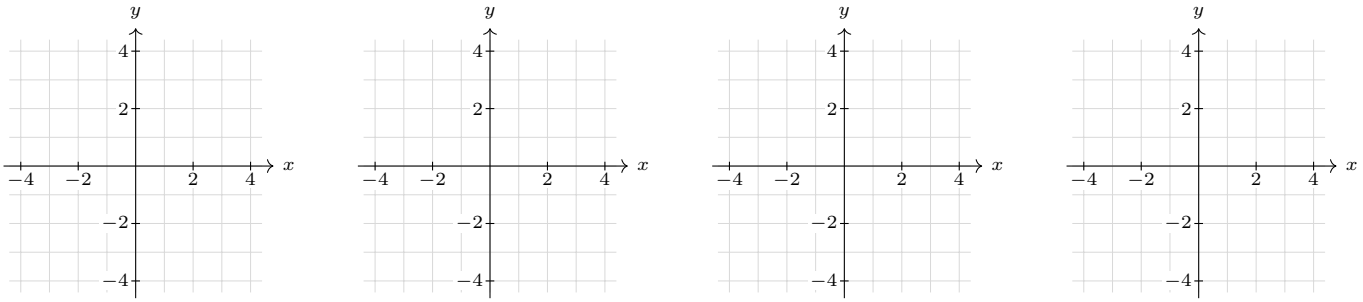
6. On each grid, sketch a line with the given characteristics.

(a) Exactly two intercepts, both positive.

(b) Exactly one intercept, which is an x-intercept.

(c) Exactly one intercept, and a positive slope.

(d) A y-intercept of -3 and a slope of 0.



7. Priya cycles home from school at a steady speed. The table shows her distance from home, d (km), t minutes after she leaves.

t (min)	0	10	20	30
d (km)	12	9	6	3

(a) Show that the relation is linear.

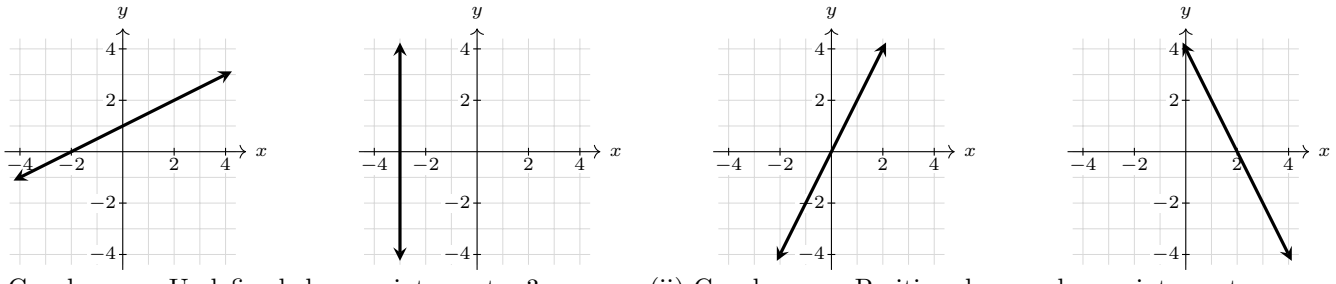
(b) Determine the slope, with units, and explain what it means.

(c) Determine both intercepts and explain what each means.

(d) State the domain and range for her trip home.

8. Match each description with its graph.

Graph P **Graph Q** **Graph R** **Graph S**



(i) Graph _____ Undefined slope; x-intercept -3.

(ii) Graph _____ Positive slope; only one intercept.

(iii) Graph _____ Negative slope; x-intercept 2, y-intercept 4.

(iv) Graph _____ Positive slope; x-intercept -2, y-intercept 1.

9. *(Multiple Choice)* Which table of values represents a linear relation?

A.

x	0	1	2	3
y	1	2	4	8

B.

x	0	1	2	3
y	5	3	1	-1

C.

x	0	1	2	3
y	0	1	4	9

D.

x	0	2	3	5
y	1	3	5	7

10. *(Multiple Choice)* A line has exactly one intercept, and its slope is negative. Which statement must be true?

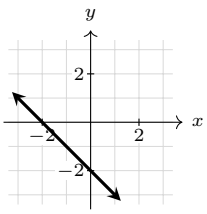
A. The line is horizontal.
B. The line is vertical.
C. The line passes through the origin.
D. The line is one of the axes.

11. *(Multiple Choice)* The domain and range of the horizontal line through $(2, -5)$ are

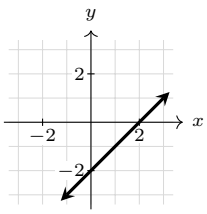
A. domain $\{2\}$, range $\{y \mid y \in \mathbb{R}\}$
B. domain $\{x \mid x \in \mathbb{R}\}$, range $\{-5\}$
C. domain $\{x \mid x \in \mathbb{R}\}$, range $\{y \mid y \in \mathbb{R}\}$
D. domain $\{2\}$, range $\{-5\}$

12. *(Multiple Choice)* Which graph shows a line with a negative slope and a positive x-intercept?

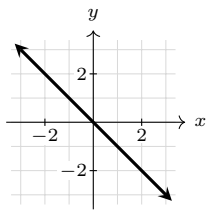
A.



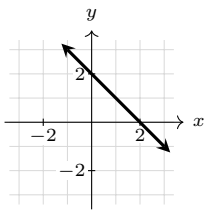
B.



C.



D.



13. *(Numerical Response)* The table represents a linear relation.

x	2	5	8
y	10	19	28

The y-intercept of its graph is _____.

Record your answer left to right. | | | | |

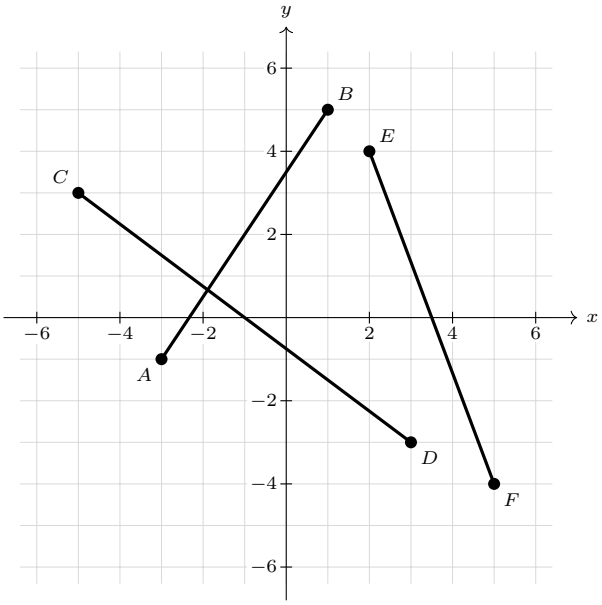
14. *(Numerical Response)* A generator burns fuel at a constant rate. After 2 h its tank holds 52 L, and after 5 h it holds 31 L. To the nearest tenth, the number of hours after starting until the tank is empty (the horizontal intercept of the graph) is _____.

Record your answer left to right. | | | | |

Lesson 5 Assignment *Show your work.*

Questions 1–13 review slope and linear graphs (Lessons 2–4). Questions 14–22, at the end, review distance and midpoint (Lesson 1).

1. *(Matching)* Three line segments are drawn on the grid below. Match each segment with its slope by writing the slope's number in the blank. Each slope may be used once, more than once, or not at all.



Slopes

[1] $-\frac{8}{3}$ [2] $\frac{3}{2}$ [3] $-\frac{3}{4}$

[4] $\frac{4}{3}$ [5] $-\frac{3}{2}$ [6] $\frac{3}{8}$

AB: _____ CD: _____ EF: _____

2. *(Multiple Choice)* Line segment AB joins $A(-3, 6)$ and $B(3, -9)$; line segment PQ joins $P(8, 2)$ and $Q(-4, -6)$. Which of the following statements is correct?

A. The slope of AB is positive and the slope of PQ is negative.
B. The slope of AB is negative and the slope of PQ is positive.
C. AB is parallel to PQ.
D. AB is perpendicular to PQ.

3. *(Multiple Choice)* K and L are the points $(6, 9)$ and $(-2, -5)$ respectively. The slope of a line perpendicular to KL is

A. $\frac{4}{7}$ B. $-\frac{4}{7}$ C. $\frac{7}{4}$ D. $-\frac{7}{4}$

4. *(Multiple Choice)* Quadrilateral $PQRS$ has vertices $P(-6, -9)$, $Q(-4, -3)$, $R(5, -6)$, and $S(3, -12)$. Consider the following statements.

I. PQ is parallel to SR. II. QR is perpendicular to SR. III. QR is parallel to PS.

Which of the following is correct?

A. Statement I is false.
B. Statement II is false.
C. Statement III is false.
D. None of the statements is false.

5. *(Multiple Choice)* A jogger's distance–time graph is a straight line through $(0, 0)$ and $(25, 4)$, where time is in minutes and distance is in kilometres. The slope of the graph is

A. 6.25 km/min B. 4 km/min C. 0.16 km/min D. 0.16 min/km

6. *(Multiple Choice)* Which slope gives the steepest line?

A. $m = -3$ B. $m = \frac{5}{2}$ C. $m = 2.9$ D. $m = -\frac{1}{4}$

7. *(Multiple Choice)* Which statement describes the relation in this table?

x	1	3	4	7
y	2	6	8	14

A. Not linear, because the x-values do not go up in equal steps.
B. Not linear, because the changes in y are 4, 2 and 6.
C. Linear, with slope 4.
D. Linear, with slope 2.

8. *(Multiple Choice)* Which line has exactly one intercept?

A. The line through $(0, 3)$ and $(3, 0)$.
B. The line through $(-2, -4)$ and $(1, 2)$.
C. The x-axis.
D. The line through $(2, 1)$ and $(4, 5)$.

9. *(Multiple Choice)* The domain and range of the vertical line through $(-3, 4)$ are

A. domain $\{x \mid x \in \mathbb{R}\}$, range $\{4\}$
B. domain $\{-3\}$, range $\{4\}$
C. domain $\{-3\}$, range $\{y \mid y \in \mathbb{R}\}$
D. domain $\{x \mid x \in \mathbb{R}\}$, range $\{y \mid y \in \mathbb{R}\}$

10. *(Numerical Response)* Two lines have slopes of $-\frac{3}{4}$ and $\frac{20}{t}$ respectively. If the lines are perpendicular, the value of t must be _____.

Record your answer left to right. | | | | |

11. *(Numerical Response)* The segment joining $K(-3, y)$ and $L(-5, 4)$ is parallel to the segment joining $M(6, 2)$ and $N(1, 5)$. The value of y, to the nearest hundredth, is _____.

Record your answer left to right. | | | | |

12. *(Numerical Response)* The points $A(-3,-7)$, $B(1,-1)$ and $C(9,k)$ are collinear. The value of k is _____.

Record your answer left to right. | | | | |

13. *(Numerical Response)* The table represents a linear relation.

x	0	4	8
y	20	14	8

To the nearest tenth, the x-intercept of its graph is _____.

Record your answer left to right. | | | | |

Written Response — 5 marks

Consider the points $A(-8,10)$, $B(1,1)$, $C(4,3)$, and $D(10,-8)$.

(a) Three of these points lie on the same straight line and one does not. **Explain** how you could algebraically determine which point is not collinear with the other three.

(b) Determine algebraically which of the points A, B, C, D is not collinear with the other three.

(c) A fifth point, $E(k,-17)$, lies on the same line as the three collinear points. **Determine** the value of k.

Distance and Midpoint

These questions use the distance and midpoint formulas from Lesson 1.

14. *(Multiple Choice)* The exact distance between the points $(1, 6)$ and $(9, -9)$ is

A. 12.53 B. $\sqrt{161}$ C. 17 D. 289

15. *(Multiple Choice) Use the following information to answer this question and question 20.* The points $A(6, 1)$ and $B(-4, 7)$ are on the circumference of a circle, and the line segment AB passes through the centre of the circle. The centre of the circle is the point

A. $(1, 4)$ B. $(2, 8)$ C. $(-1, 4)$ D. $(5, -3)$

16. *(Multiple Choice)* Consider AB joining $A(7, -5)$ and $B(-3, -5)$, and CD joining $C(2, -10)$ and $D(2, 0)$. Which one of the following statements about these line segments is true?

A. AB and CD have the same slope and are equal in length.
B. AB and CD have the same slope and are unequal in length.
C. CD has a length of 10 units and a slope of zero.
D. AB and CD have the same midpoint and are equal in length.

17. *(Multiple Choice)* Line segment AB joins $A(-3, 6)$ and $B(3, -9)$; line segment PQ joins $P(8, 2)$ and $Q(-4, -6)$. Which of the following statements is correct?

A. The midpoint of AB has an x-coordinate greater than the midpoint of PQ.
B. The midpoint of AB has a y-coordinate greater than the midpoint of PQ.
C. The midpoints of AB and PQ are the same point.
D. The line segment joining the midpoints is horizontal.

18. *(Multiple Choice)* Which of the following most completely describes quadrilateral $PQRS$ from question 4?

A. rectangle B. square C. rhombus D. parallelogram

19. *(Multiple Choice)* Triangle GHJ has vertices $G(-5, 2)$, $H(1, -6)$, and $J(7, 2)$. Which of the following most completely describes $\triangle GHJ$?

A. scalene, with no right angle
B. isosceles, with no right angle
C. isosceles, with a right angle
D. equilateral

20. *(Numerical Response)* The points $A(6, 1)$ and $B(-4, 7)$ from question 15 lie on a circle, with AB passing through the centre. The area of the circle, to the nearest whole number, is _____.

Record your answer left to right. | | | | |

21. *(Numerical Response)* PQ is the diameter of a circle whose centre is R. If $Q(5.4, 3.2)$ and $R(9.1, 7.9)$, then the x-coordinate of P is _____.

Record your answer left to right. | | | | |

22. Using the points of the Written Response, determine algebraically, without using the midpoint formula, that $B(1, 1)$ is the midpoint of line segment AD.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) 6 (b) 11 (c) 42 (d) 2029. **2.** 60 units^2. **3.** $AB = 2\sqrt{17} \approx 8.25$; $CD = \sqrt{61} \approx 7.81$; $EF = 4\sqrt{10} \approx 12.65$. **4.** (a) 10 (b) 13 (c) $2\sqrt{10}$ (d) 10.63. **5.** (a) Maria $(5, 8)$, Nathan $(-7, -4)$ (b) ≈ 1867 m. **6.** (a) $(6, 9)$ (b) $(-8, -3)$ (c) $(1, -4)$ (d) $(-0.5, -1.8)$. **7.** (a) -3 (b) -5. **8.** (a) $(2855, 2135)$ (b) ≈ 75 km. **9.** D. **10.** A. **11.** 25.6. **12.** 9.

Lesson 2 Assignment

1. AB: rise 4, run 8, slope $\frac{1}{2}$; CD: rise -4, run 6, slope $-\frac{2}{3}$; EF: rise -6, run 4, slope $-\frac{3}{2}$; GH: rise 6, run 6, slope 1. **2.** Line $1 = \frac{1}{2}$, Line $2 = 4$, Line $3 = -\frac{1}{3}$, Line $4 = -3$. **3.** Any three of $(-3, -1)$, $(0, 1)$, $(3, 3)$, $(6, 5)$. **4.** (a) $(8, 2)$, $(-2, -4)$, $(-7, -7)$ (b) $(9, 1)$, $(1, 3)$, $(-3, 4)$ (c) $(-4, 2)$, $(-2, -3)$, $(0, -8)$ (d) $(1, -2)$, $(2, 1)$, $(3, 4)$ (e) $(0, -5)$, $(-1, -2)$, $(-2, 1)$ (f) $(-5, 7)$, $(3, 7)$, $(6, 7)$ (other correct points accepted). **5.** Many answers; any two of (a) $(3, 0)$, $(4, 3)$, $(5, 6)$ (b) $(3, -5)$, $(4, -7)$, $(5, -9)$ (c) $(7, -1)$, $(12, 1)$, $(-3, -5)$ (d) $(6, -6)$, $(10, -9)$, $(-2, 0)$ (e) $(5, -3)$, $(-2, -3)$, $(10, -3)$ (f) $(2, 4)$, $(2, -10)$, $(2, 0)$. **6.** (a) rise $= 15$ (b) run $= 28$ (c) rise $= -15$ (d) run $= 30$. **7.** 1.5 m. **8.** 5.4 cm^2. **9.** (a) 18 km/h (b) she rides 18 km every hour (her speed). **10.** (a) -6 dollars per week (b) the balance drops \$6 each week (c) 10 weeks. **11.** (a) $\frac{2}{25}$ (b) 120 m. **12.** (a) No: $\frac{45}{480} = \frac{3}{32} \approx 0.094 > \frac{1}{12} \approx 0.083$, too steep (b) 5.4 m. **13.** D. **14.** A. **15.** C. **16.** B. **17.** 2.4. **18.** 1.7.

Lesson 3 Assignment

1. Segment 1 positive, Segment 2 negative, Segment 3 zero, Segment 4 undefined, Segment 5 negative, Segment 6 positive. **2.** (a) $-\frac{2}{3}$ (b) -1 (c) $-\frac{3}{4}$ (d) $-\frac{9}{11}$ (e) undefined (f) 0. **3.** (a) $\frac{7}{2}$ (b) $(11, 36.5)$ (c) 65.5 m. **4.** (a) $k = 11$ (b) $k = -10$ (c) $k = 12$. **5.** (a) $m_{XY} = -\frac{3}{7} = m_{YZ}$, shared point Y, so collinear (b) $XY = \sqrt{58}$, $YZ = 2\sqrt{58}$, $XZ = 3\sqrt{58}$; $XY + YZ = XZ$. **6.** (a) $m_{AB} = \frac{1}{2} = m_{BC}$, so collinear (b) no; $m_{AD} = \frac{5}{8} \neq \frac{1}{2} = m_{AC}$ (c) $k = 8$. **7.** Only the boat crew reaches P: truck $m = -\frac{2}{5}$ but $m_{A_2P} = -\frac{1}{6}$; boat $m = -\frac{3}{2} = m_{B_2P}$. **8.** C. **9.** B. **10.** A. **11.** 1.4. **12.** (a) parallel (b) perpendicular (c) parallel (d) neither (e) perpendicular (f) neither. **13.** Parallel: $AB \parallel GH$, $IJ \parallel PQ$; perpendicular: $AB \perp CD$, $GH \perp CD$, $EF \perp IJ$, $EF \perp PQ$, $KL \perp MN$. **14.** Yes: one segment is vertical and the other horizontal. **15.** (a) $\frac{4}{3}$ (b) $-\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{3}{2}$. **16.** (a) not right-angled (b) right-angled at B. **17.** (a) $k = 20$ (b) $n = -2$ (c) $m = \frac{7}{36}$ (d) $w = -10$. **18.** (a) $h = -1$ (b) $p = -72$ (c) $s = \frac{3}{4}$ (d) $q = -25$. **19.** $m_{QS} = -\frac{3}{2}$. **20.** (a) $m_{AB} = \frac{5}{4}$, $m_{BC} = -\frac{4}{5}$, product -1 (b) $AB^2 = 41$, $BC^2 = 41$, $AC^2 = 82 = 41 + 41$. **21.** D. **22.** $p = -1.5$. **23.** (a) he subtracted the y's as $N - M$ but the x's as $M - N$ (b) $m_{MN} = -2$; perpendicular slope $\frac{1}{2}$. **24.** 2.4. **25.** (a) $k = -1$ (b) $t = -5$.

Lesson 4 Assignment

1. (a) linear, slope 3 (b) not linear (c) linear, slope 0 (d) linear, slope -2. **2.** (a) 1, -2, -8 (b) -1, 2, 4. **3.** (a) x-int -4, y-int -2, slope $-\frac{1}{2}$ (b) x-int -1, y-int 3, slope 3 (c) x-int 2, no y-int, slope undefined. **4.** (a) x-int 0, y-int 0 (b) x-int 3, y-int 6. **5.** (a) D: $\{x \mid x \in \mathbb{R}\}$, R: $\{y \mid y \in \mathbb{R}\}$ (b) D: $\{x \mid x \in \mathbb{R}\}$, R: $\{5\}$ (c) D: $\{4\}$, R: $\{y \mid y \in \mathbb{R}\}$ (d) D: $\{x \mid -2 \leq x \leq 3,\ x \in \mathbb{R}\}$, R: $\{y \mid -3 \leq y \leq 7,\ y \in \mathbb{R}\}$. **6.** Sample answers: (a) through $(3, 0)$ and $(0, 2)$ (b) $x = 3$ (c) through $(0, 0)$ and $(1, 1)$ (d) $y = -3$. **7.** (a) d drops 3 every 10 min (b) -0.3 km/min; she gets 0.3 km closer each minute (c) d-int 12: starts 12 km away; t-int 40: home after 40 min (d) D: $\{t \mid 0 \leq t \leq 40,\ t \in \mathbb{R}\}$, R: $\{d \mid 0 \leq d \leq 12,\ d \in \mathbb{R}\}$. **8.** (i) Q (ii) R (iii) S (iv) P. **9.** B. **10.** C. **11.** B. **12.** D. **13.** 4. **14.** 9.4.

Lesson 5 Assignment

1. $AB \to 2$, $CD \to 3$, $EF \to 1$. **2.** B **3.** B **4.** D **5.** C **6.** A **7.** D **8.** B **9.** C **10.** 15 **11.** 2.80 **12.** 11 **13.** 13.3
Written Response: (a) Find the slope from one fixed point to each of the other three; two equal slopes identify the line, and the point giving the odd slope is off it (if all three differ, the fixed point is the odd one). (b) $m_{AB} = -1$, $m_{AD} = -1$, $m_{AC} = -\frac{7}{12}$, so A, B, D are collinear and C is not. (c) $k = 19$. **Distance and midpoint: 14.** C **15.** A **16.** D **17.** B **18.** A **19.** B **20.** 107 **21.** 12.8 **22.** $AB = BD = 9\sqrt{2}$ and A, B, D are collinear, so B is the midpoint of AD.